# Personal Meeting

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## 1 HRSID Pixel Scale

The echo simulation uses each HRSID pixel value directly as a scatterer amplitude, which requires two properties of the pixel value to be confirmed: (1) linear, not logarithmic; (2) amplitude, not amplitude squared.

**Product specifications.**

- HRSID is cropped from Sentinel-1B, TerraSAR-X and TanDEM-X panoramas [1].
- Sentinel-1 Level-1 GRD: $\beta^0 = DN^2/A_\beta^2$, so the stored DN is amplitude [2].
- TerraSAR-X/TanDEM-X detected products: $\beta^0 = k_s\, DN^2$, so DN is amplitude [3].
- How HRSID converted the products to 8-bit images is not documented.

**Data.** 975,864 homogeneous $32 \times 32$ sea patches from 1593 offshore scenes, each compared with synthetic speckle at the same mean levels, integer rounding and JPEG compression.

**Test 1: linear vs. log.** Let $x$ be the grey value of one pixel in a homogeneous sea patch. The complex return of a pixel is a sum of many scatterers with random phases, a zero-mean complex Gaussian $s \sim \mathcal{CN}(0, \sigma^2)$, so $s = \sigma u$ with $u \sim \mathcal{CN}(0, 1)$: the backscatter only sets the scale $\sigma$. Taking the magnitude keeps the scale out, $|\sigma u| = \sigma|u|$ and $|\sigma u|^2 = \sigma^2|u|^2$, so in linear units

$$x = c\, n, \qquad \begin{cases} \text{amplitude: } c = \sigma, \quad n = |u| \sim \text{Rayleigh (Nakagami for } L \text{ looks)}, \\ \text{intensity: } c = \sigma^2, \ n = |u|^2 \sim \text{exponential (Gamma for } L \text{ looks)}, \end{cases}$$

where the distribution of $n$ does not depend on $c$. Within a patch $c$ is a constant, so the mean and the standard deviation scale with $c$:

$$\operatorname{mean}(x) = E[c\, n] = c\, E[n],$$
$$\operatorname{Var}(x) = E\big[(c\, n - c\, E[n])^2\big] = c^2 \operatorname{Var}(n) \ \Rightarrow\ \operatorname{std}(x) = c \operatorname{std}(n) \quad (c > 0).$$

In both cases $c$ cancels in the ratio

$$\frac{\operatorname{std}(x)}{\operatorname{mean}(x)} = \frac{c \operatorname{std}(n)}{c\, E[n]} = \frac{\operatorname{std}(n)}{E[n]} = \mathrm{CV} \ \Rightarrow\ \log \operatorname{std}(x) = \log \operatorname{mean}(x) + \log \mathrm{CV},$$

with the coefficient of variation CV fixed by the distribution (about 0.52 for Rayleigh, 1 for exponential, $1/\sqrt{L}$ for Gamma), so the slope of log(std) vs. log(mean) over the patches is $k = 1$ whether the pixel is amplitude or intensity. After a log mapping, $x = \alpha \log(c\, n) + \beta = \alpha \log c + \alpha \log n + \beta$. Within a patch $c$ is a constant, so

$$\operatorname{mean}(x) = \alpha \log c + \alpha\, E[\log n] + \beta,$$
$$\operatorname{Var}(x) = E\big[(\alpha \log n - \alpha\, E[\log n])^2\big] = \alpha^2 \operatorname{Var}(\log n) \ \Rightarrow\ \operatorname{std}(x) = \alpha \operatorname{std}(\log n) \quad (\alpha > 0).$$

The mean moves with $c$ but the standard deviation does not, so $\log \operatorname{std}(x) = \text{const}$ over the patches and $k = 0$.

Table 1: Test 1: slope $k$ and intercept $b$ of $\log \operatorname{std}(x) = k \log \operatorname{mean}(x) + b$.

| | slope $k$ | intercept $b$ |
|---|---|---|
| linear (amplitude or intensity), theory | 1 | $\log \mathrm{CV}$ |
| log, theory | 0 | $\log(\alpha \operatorname{std}(\log n))$ |
| HRSID sea (fit) | 1.17 | $-1.17$ |

**Test 2: amplitude vs. intensity.** Assumes linear pixel values (Test 1). For $L$-look speckle the number of looks $L$ is estimated twice from the same patches, from $E[x^2]/E[x]^2$ and from $\mathrm{Var}(\ln x)$; the two estimates agree only under the right hypothesis.

Table 2: Test 2: two estimates of $L$ under each hypothesis.

| | $L$, amplitude hypothesis | $L$, intensity hypothesis |
|---|---|---|
| HRSID sea | 1.10 / 1.15 | 4.03 / 3.46 |
| synthetic 1-look amplitude | 0.91 / 0.97 | 3.32 / 2.78 |
| synthetic 4-look intensity | 0.98 / 1.08 | 3.60 / 3.19 |

**Conclusion.**

- Linear: confirmed ($k = 1.17$, far from the dB range).
- Amplitude: favoured but not proven. The two $L$ estimates agree within 4 % under amplitude and 15 % under intensity, and amplitude is more consistent in 113 of 123 panoramas; 4-look intensity is hard to tell from 1-look amplitude.

# References


[1] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020.

[2] ESA, "Radiometric calibration of Sentinel-1 Level-1 products," Sentinel Online. [Online]. Available: `https://sentinels.copernicus.eu/web/sentinel/radiometric-calibration-of-level-1-products`

[3] Airbus Defence and Space, "Radiometric calculations for TerraSAR-X data," Tech. Note TSXX-ITD-TN-0049.